HYDRODYNAMIC SINGULARITIES

When fluids forget their past

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What is Singularity?

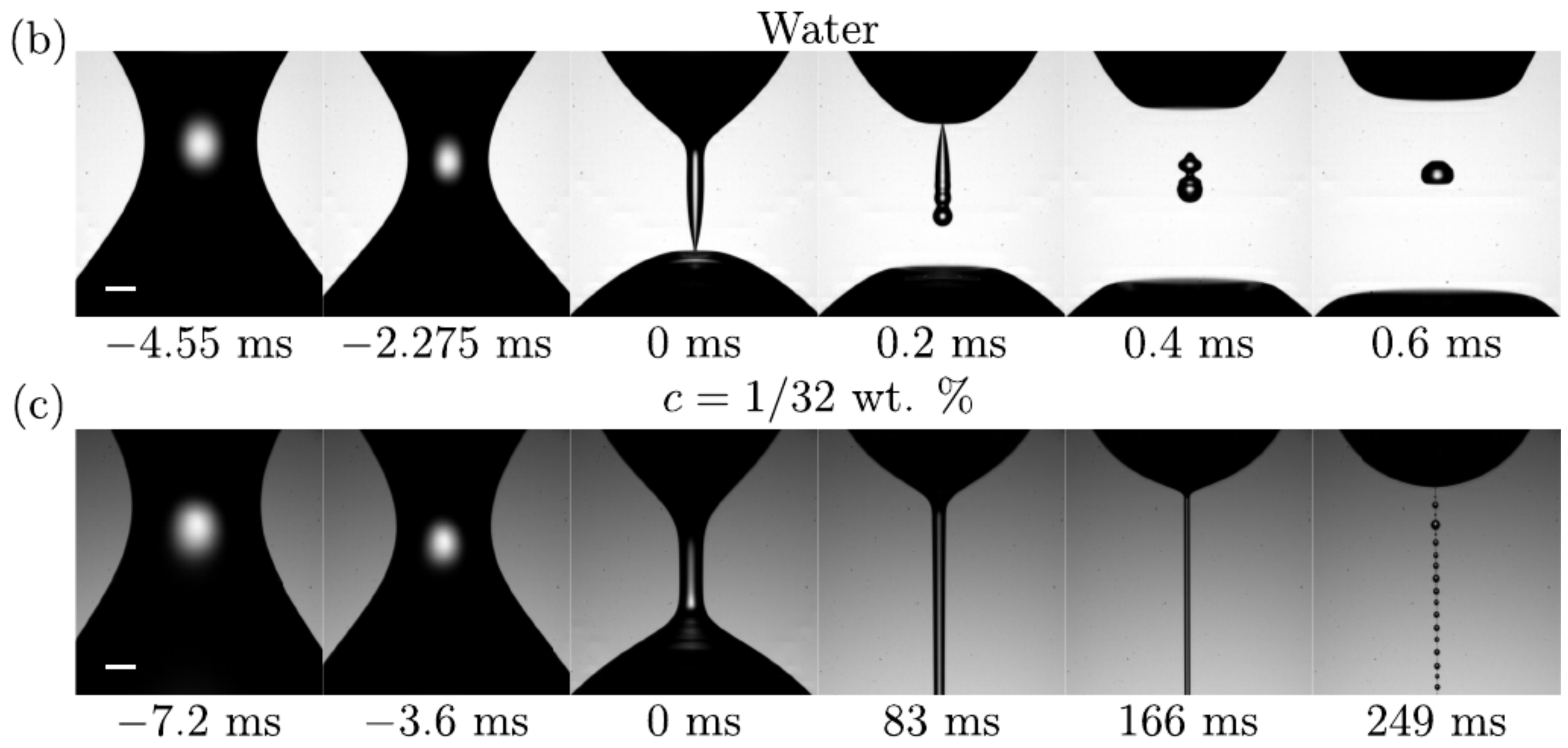
A singularity is when a smooth physical process "blows up" - something finite becomes infinite in a finite time or at a finite place.

Close to a singularity, the system loses memory of its initial condition: the local dynamics become universal and self-similar.

1. DROPS: ELASTICITY CHANGES THE ROUTE

Drop pinch-off

Newtonian liquid: the neck radius collapses to a true finite-time singularity.



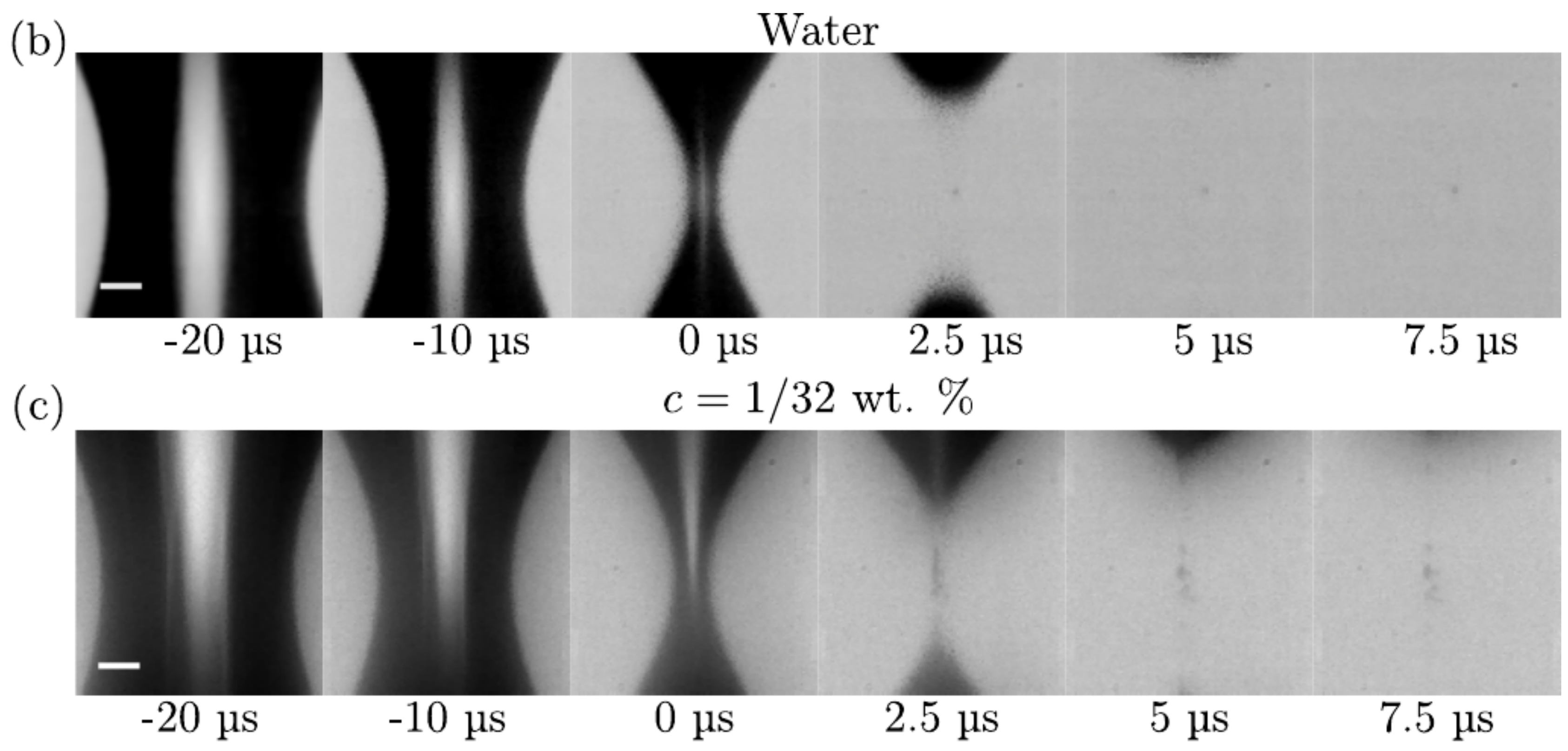
Add polymers: the singularity is arrested; the drop forms a long thread and beads-on-a-string instead.

polymers kill the singularity

2. BUBBLES: ELASTICITY LEAVES THE SINGULARITY INTACT

Bubble pinch-off

Newtonian liquid: the neck pinches to a singular point.

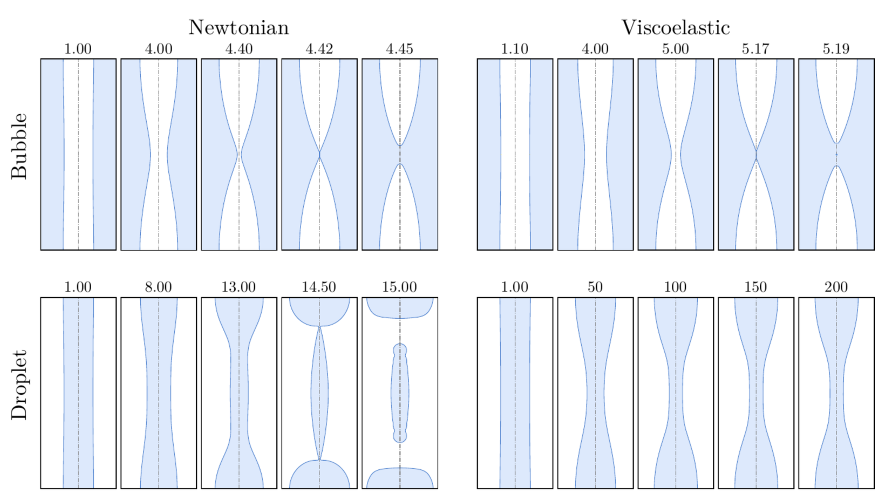


Add polymers: the near-singular sequence looks almost unchanged. The singularity survives.

polymers do not erase it

3. WHY SINGULARITIES MATTER

Numerical snapshots: controlled comparison



The simulations make the contrast explicit: droplets are regularised by elasticity into persistent filaments, while bubbles retain a Newtonian-looking singular pinch-off pathway.

Scaling-law blurb

near pinch-off: h_min ~ (t0 - t)^alpha

self-similarity: shapes collapse after rescaling by the local neck radius

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